

**eFlow: A Web and Mobile-Based e-Governance Project and Task
Management System for the Local Economic Development and
Investment Promotion Office with Artificial Intelligence Integration**

**A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Ormoc**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

**Ernst Livin Gabriel B. Cahiyang
James Adrian A. Denoy
Kurt Russell C. Dichosa
Andres U. Manili**

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration**

NAME OF PROPONENTS: Ernst Livin Gabriel B. Cahiyang
James Adrian A. Denoy
Kurt Russell C. Dichosa
Andres U. Manili

In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

ENDORSED BY:

Rona Mira B. Lucañas
Capstone Project Adviser

APPROVED FOR PROPOSAL DEFENSE:

Rona Mira B. Lucañas
Capstone Project Coordinator

NOTED BY:

Rona Mira B. Lucañas
Program Head

May 11, 2026
APPROVAL SHEET

This capstone project proposal titled of **eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration**, prepared and submitted by **Ernst Livin Gabriel B. Cahiyang, James Adrian A. Denoy, Kurt Russell C. Dichosa, Andres U. Manili**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

Rona Mira B. Lucañas
Capstone Project Adviser

Accepted and approved by the Capstone Project Review Panel
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

Rona Mira B. Lucañas
Panel Member

Elizabeth Dumaran
Panel Member

Engr. Sheena Joy S. Muyuela
Lead Panelist

Noted:

Rona Mira B. Lucañas
Capstone Project Coordinator

Rona Mira B. Lucañas
Program Head

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INTRODUCTION

Project Context

The rapid advancement of digital technologies has significantly influenced how organizations operate, particularly within government offices. Public institutions are increasingly expected to transition from traditional, paper-based processes to digital platforms that promote efficiency, transparency, and accountability. In the Philippines, this transformation is further encouraged by national initiatives such as the Ease of Doing Business Act, which mandates government agencies to simplify processes and improve service delivery through digitalization. In Ormoc City, various government offices have embraced modernization through the adoption of digital governance initiatives. The city has implemented programs such as the Electronic Business One-Stop Shop (eBOSS), which allows business-related transactions and permit processing to be conducted more efficiently through an online platform. These initiatives have contributed to improved accessibility, convenience, and efficiency in government service delivery. Despite these advancements, operational challenges remain evident within the Local Economic Development and Investment Promotion Office (LEDIPO) of Ormoc City, particularly in the management of projects, programs, and assigned tasks. Currently, project monitoring and task tracking are often carried out using fragmented systems and manual processes. Microsoft Excel or Google Spreadsheets are commonly utilized to record project updates and monitor progress, while task assignments are typically communicated manually by supervisors or department heads. Meetings are conducted either face-to-face or through Zoom, and employee communication is often facilitated through text messages or Viber. While these tools provide temporary support for daily operations, the absence of an

integrated project and task management platform frequently results in delayed updates, duplication of work, difficulty in monitoring project status, miscommunication, and increased administrative workload. Employees may also encounter challenges in accessing project-related information because data and communications are dispersed across multiple platforms. Furthermore, varying levels of digital literacy among personnel, resistance to technological change, and occasional internet connectivity issues can hinder the effective adoption of digital solutions. Although numerous commercial project and task management systems are available, many are not specifically designed to address the unique operational requirements of government offices such as LEDIPO. These systems may not fully align with public sector workflows, reporting requirements, and the organizational environment in which local government offices operate. Given these challenges, there is a need for a customized solution that integrates project and task management into a unified digital platform tailored to the operational needs of LEDIPO. This study proposes the development of **eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office (LEDIPO) of Ormoc City with Artificial Intelligence Integration**. The proposed system aims to streamline workflows, improve coordination, centralize communication, enhance project monitoring, and increase overall efficiency and convenience for employees and administrators. By addressing existing gaps in project and task management practices, the system supports the ongoing digital transformation efforts of the office and contributes to more effective public service delivery.

Purpose and Description

The purpose of this project is to design and develop a web and mobile-based system that will help the Local Economic Development and Investment Promotion Office (LEDIPO) of Ormoc City manage projects, programs, and tasks more efficiently. The system aims to replace manual and fragmented processes with a centralized platform where users can create, assign, monitor, and update tasks in real time. The proposed system, eFlow, will provide features such as task assignment, progress tracking, notifications, reporting, and collaboration tools. It is designed to be user-friendly and accessible through both web browsers and mobile devices, making it convenient for employees to use regardless of their location. The system will also serve as a centralized repository of project-related information, enabling users to access updates, documents, and communications within a single platform. What makes this project unique is its focus on the specific operational needs of LEDIPO. Unlike generic task management systems, eFlow is tailored to the office's workflows, taking into consideration project monitoring requirements, approval processes, reporting needs, and varying levels of user expertise. It also addresses common challenges such as delayed updates, scattered communication channels, and reliance on physical meetings by enabling real-time communication, task monitoring, and collaboration among employees. Furthermore, the integration of Artificial Intelligence (AI) will support smarter project and task management by providing automated notifications, task prioritization assistance, and data-driven insights that can help users make informed decisions. Through these features, the system seeks to improve coordination, enhance productivity, reduce delays, and support more efficient management of projects and activities within LEDIPO. Overall, eFlow aims to strengthen the office's operational

efficiency by providing a customized digital solution that supports its ongoing modernization efforts and contributes to the delivery of more effective services for economic development and investment promotion in Ormoc City.

Objectives

General Objective:

To develop eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office (LEDIPO) with Artificial Intelligence Integration, that centralizes project coordination, task allocation, workflow monitoring, communication, and intelligent decision-support processes in order to improve operational efficiency, transparency, accountability, and digital governance within the Local Economic Development and Investment Promotion Office (LEDIPO).

Specific Objectives:

- **To analyze the existing project and task management processes, workflow procedures, communication methods, and operational challenges within LEDIPO.**

This identifies inefficiencies, delays, fragmented communication, workload imbalance, and limitations of traditional manual systems to determine the functional and technical requirements of the proposed platform.

- **To design a centralized web and mobile-based e-governance project and task management system architecture for administrators, supervisors, and employees.**

This includes designing the system interface, database structure, workflow processes, security mechanisms, AI-assisted modules, blockchain audit trail integration, and user interaction features to ensure an organized, scalable, and user-friendly platform.

- **To develop a centralized platform that enables administrators, supervisors, and employees to create, assign, monitor, and manage projects and tasks within LEDIPO.**

This ensures that all project-related activities, workflows, and records are organized within a single integrated system, reducing fragmented processes and improving interdepartmental coordination.

- **To implement a real-time task monitoring and progress tracking feature for employees and supervisors.**

This allows users to continuously monitor project progress, task completion status, delays, and bottlenecks, improving accountability, workflow visibility, and operational efficiency.

- **To develop a notification and alert system for task assignments, project updates, deadlines, announcements, and workflow changes.**

This ensures that users receive timely updates and reminders regarding their responsibilities, minimizing communication delays and reducing the possibility of missed deadlines.

- **To provide a reporting and analytics module capable of generating project summaries, task completion reports, productivity analytics, and departmental performance insights.**

This supports data-driven decision-making by enabling administrators and supervisors to evaluate employee productivity, monitor project performance, and analyze workflow efficiency.

- **To implement an Artificial Intelligence (AI)-powered workload allocation feature that assists supervisors in distributing tasks efficiently based on employee availability, workload capacity, task priority, and project deadlines.**

This improves resource utilization, promotes balanced workload distribution, and helps prevent task overload among employees.

- **To integrate a Genetic Algorithm optimization mechanism for intelligent task scheduling and workload balancing.**

This enables the system to automatically optimize task assignments and scheduling decisions by analyzing operational constraints, workload conditions, and personnel capabilities to improve overall efficiency.

- **To implement Process Mining functionalities for workflow analysis and operational monitoring within LEDIPO.**

This allows the system to analyze workflow patterns, identify inefficiencies and process bottlenecks, and provide insights that support process improvement and organizational optimization.

- **To develop a blockchain-based audit trail feature that records task-related actions, approvals, updates, and system transactions in an immutable and transparent manner.**

This ensures data integrity, accountability, traceability, and protection against unauthorized modification of records within the system.

- **To create a user-friendly web and mobile interface that accommodates users with varying levels of digital literacy.**

This improves accessibility, usability, and user adoption by ensuring that employees and administrators can easily navigate and utilize the system regardless of technical experience.

- **To implement a secure authentication and role-based access control system for administrators, supervisors, and employees.**

This ensures that only authorized users can access specific modules and information, helping maintain system security, confidentiality, and proper management of organizational data.

- **To enable employees and supervisors to communicate and collaborate through task updates, comments, notifications, and shared project activities within the platform.**

This strengthens coordination between departments and minimizes reliance on traditional manual communication methods and frequent face-to-face meetings.

- **To develop a centralized database system for securely storing and managing project records, task information, workflow logs, notifications, reports, audit trails, and user accounts.**

This improves data accessibility, consistency, organization, and security while reducing redundancy and manual record management.

- **To implement the proposed eFlow system within the operational environment of LEDIPO through web and mobile deployment.**

This ensures that authorized users can access and utilize the platform for project management, task coordination, workflow monitoring, AI-assisted scheduling, and secure digital collaboration across departments.

- **To evaluate the functionality, usability, efficiency, security, and performance of the proposed system based on user testing and system assessment criteria.**

This determines whether the system effectively improves workflow management, task monitoring, coordination, transparency, and operational efficiency within LEDIPO.

- **To gather feedback and recommendations from administrators, supervisors, employees, and system users regarding the effectiveness and usability of the proposed platform.**

This helps identify areas for future enhancement, system optimization, feature improvement, and continuous development to better support the digital transformation initiatives of LEDIPO.

- **To support the digital transformation initiatives of LEDIPO by replacing traditional manual workflow management processes with an intelligent, secure, and technology-driven e-governance platform.**

This promotes transparency, accountability, productivity, operational efficiency, and modernized public administration through the integration of advanced digital technologies.

Scope and Limitations

User Levels

- **Administrator** – Has full control over the system, including managing users, creating and updating projects, assigning tasks, monitoring all departmental activities, generating reports, and managing system configurations. The administrator also oversees system-wide performance and ensures proper data security and integrity.
- **Department Head / Supervisor** – Responsible for managing projects within their assigned department. This includes creating and assigning tasks to employees, setting deadlines, monitoring task progress, evaluating employee performance, and generating departmental reports for analysis and decision-making.
- **Employee / User** – Serves as the end-user of the system. Employees can view assigned tasks, update task progress, submit reports or outputs, receive notifications, and communicate updates related to their assigned responsibilities within the system.

Platform Accessibility

The system is accessible through both web and mobile platforms, allowing users to manage and monitor tasks using computers, laptops, tablets, and mobile devices. Both platforms are designed to provide a consistent user experience, ensuring that all core features such as task management, notifications, reporting, and monitoring are fully functional across devices. Additionally, both platforms are connected to a centralized database that enables real-time synchronization of tasks, updates, notifications, and reports. This ensures that all

users have access to accurate and updated information at all times, regardless of the device being used.

Admin (Web-Based System)

- **User Management** – Add, update, deactivate, and manage system users and assign roles and permissions.
- **Project Management** – Create, update, and oversee all projects across departments.
- **Task Oversight** – Monitor all task assignments, progress, and completion status in real time.
- **Reports Management** – Generate system-wide reports including task completion, project performance, and productivity summaries.

Department Head / Supervisor (Web-Based System)

- **Task Assignment** – Assign tasks to employees with deadlines, priorities, and instructions.
- **Project Monitoring** – Track progress of ongoing projects and evaluate task completion.
- **Performance Evaluation** – Review employee output and productivity based on task updates.
- **Report Generation** – Generate departmental reports for monitoring and decision-making.
- **Task Verification** – Review and validate completed tasks submitted by employees.

Employee / User (Web and Mobile Application)

- **Task Access** – View assigned tasks, deadlines, and project details.

- **Task Updates** – Submit progress updates, completion reports, and task outputs.
- **Notification Access** – Receive updates, reminders, and task-related announcements.
- **Communication Tools** – Provide task-related feedback and updates within the system.
- **Profile Management** – Manage personal information and account credentials.

Transactions / Processes

- **Task Assignment** – Supervisors assign tasks to employees based on project requirements.
- **Task Monitoring** – Real-time tracking of task status such as Pending, Ongoing, Completed, and Delayed.
- **Task Completion** – Employees submit completed tasks for review and validation.
- **Progress Updates** – Continuous updating of task status for accurate monitoring.
- **Reporting Process** – System generates automated reports based on task and project data.

Reports

- **Task Progress Reports** – Shows completion status, delays, and ongoing activities.
- **Project Performance Reports** – Summarizes overall project progress and outcomes.
- **Employee Productivity Reports** – Displays task completion rates and performance indicators per employee.
- **Departmental Reports** – Provides overview of departmental workload and efficiency

Utilities

- **Notification System** – Provides real-time updates for task assignments, deadlines, and changes.
- **Role-Based Access Control** – Ensures proper access depending on user roles (admin, supervisor, employee).
- **Activity Logs** – Records system actions for monitoring and auditing purposes.
- **Dashboard Analytics** – Displays visual summaries of project and task performance.

Limitations of the Study

- The system is designed strictly for internal project and task management within LEDIPO and does not include external or public-facing services such as citizen requests or public service applications.
- The system requires continuous internet connectivity for full system functionality, particularly for blockchain transactions, data synchronization, and cloud-based services. The system does not support full offline operation.
- The system is accessible through both web and mobile platforms; however, performance may vary depending on device specifications and network conditions.

Review of Related Literature/Studies/Systems

Understanding Local Government Digital Technology Adoption Strategies: A PRISMA Review

The study Understanding Local Government Digital Technology Adoption Strategies: A PRISMA Review discusses how local government units are increasingly adopting digital technologies such as cloud computing, artificial intelligence, data analytics, e-governance systems, and mobile applications to improve public service delivery. The literature highlights that digital transformation leads to faster transactions, improved transparency, better communication, and increased operational efficiency in government operations. It also emphasizes that successful adoption depends on factors such as leadership support, employee competence, organizational readiness, and proper infrastructure. However, the study also identifies challenges including limited funding, cybersecurity risks, resistance to change, poor infrastructure, and lack of technical expertise, which often hinder successful implementation of digital systems in local governments.

In relation to the purpose of this project, eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, the literature strongly supports the need for digital transformation in local government operations. Both the reviewed study and the proposed system aim to improve efficiency, reduce manual processes, and enhance coordination within government offices. However, while the literature focuses broadly on e-governance systems for public service delivery and citizen engagement, eFlow is specifically designed for internal project and task management within LEDIPO. It addresses a more focused operational gap by replacing fragmented tools

such as spreadsheets, manual tracking, and scattered communication channels with a centralized platform tailored to LEDIPO workflows.

The key findings from the literature indicate that digital systems significantly improve administrative efficiency, transparency, and decision-making through real-time data access and automation. It also reveals that mobile and cloud-based systems enhance accessibility and responsiveness in government operations. However, the literature further finds that adoption success depends heavily on user readiness, infrastructure stability, and proper training, while challenges such as resistance to change, limited digital skills, and connectivity issues remain major barriers. These findings justify the development of eFlow, as it directly addresses these gaps by proposing a user-friendly, LEDIPO-specific system enhanced with artificial intelligence features such as task automation, smart recommendations, and improved workflow support to strengthen productivity and decision-making within the local government.

Digital Transformation Adoption in the Local Government Unit (LGU) of the Science City of Muñoz, Nueva Ecija: The Challenges and Best Practices

The study Digital Transformation Adoption in the Local Government Unit (LGU) of the Science City of Muñoz, Nueva Ecija: The Challenges and Best Practices explains how digital transformation plays an important role in improving efficiency, transparency, citizen engagement, and public service delivery in local government units. The use of information and communication technology (ICT) allows LGUs to modernize their processes and provide faster and more accessible services to the public. It further explains that e-governance systems help automate common government processes such as permit processing, tax collection, document handling, and feedback management. As a result,

manual work is reduced, delays are minimized, and government records become more accurate. The study also emphasizes that digital systems improve transparency since information becomes easier to access and monitor.

In connection with eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, the literature strongly supports the need for digital systems in government operations. Both the study and eFlow aim to improve efficiency, reduce manual processes, and enhance communication within government offices. However, while the literature focuses on general e-governance systems for public services, eFlow concentrates on internal project and task management within LEDIPO. It addresses issues such as spreadsheet-based tracking, manual monitoring, and scattered communication tools by integrating them into one centralized system tailored for LEDIPO workflows. Findings from the study reveal that successful digital transformation depends on strong leadership, organizational readiness, and employee skills. It also identifies major challenges such as limited funding, poor internet connectivity, resistance to change, and low digital literacy among employees. To address these issues, the study recommends continuous training and stronger government policies for digital systems. These findings support the development of eFlow, as it provides a simple and organized platform with artificial intelligence features that enhance task management, communication, and decision-making within the LEDIPO.

Government Predictive Analytics: From Reactive Reports to Strategic Foresight

The article Government Predictive Analytics: From Reactive Reports to Strategic Foresight, explains that predictive analytics is an important tool in modern governance. It

shows how government agencies move from reactive reporting to proactive decision-making. Instead of only reviewing past reports, predictive analytics uses historical data to forecast future problems and needs. The literature explains that data such as budget reports, project timelines, service performance, and citizen feedback can be analyzed to identify trends, risks, and delays. This helps government agencies improve planning, resource use, and service delivery. The article also states that predictive analytics improves efficiency and decision-making in government operations. It allows agencies to monitor performance, predict service demands, and identify problems early. It also supports transparency and accountability because decisions are based on data and measurable results. However, the literature also identifies challenges such as poor data quality, outdated systems, fragmented records, cybersecurity risks, and lack of technical skills among employees. Resistance to change and lack of training are also major barriers to successful use of predictive systems in government.

In relation to the purpose of this project, eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, the literature supports the idea of using data-driven systems in government operations. Both the study and eFlow aim to improve decision-making, efficiency, and monitoring of tasks and projects. However, the literature focuses on predictive analytics for overall government planning, while eFlow focuses on internal project and task management within LEDIPO. eFlow also applies AI features in a more practical way, such as task tracking, notifications, and workflow support for employees. The findings of the literature show that predictive analytics improves planning, efficiency, and decision-making in government systems. It

also shows that data-driven tools help reduce delays and improve transparency. However, success depends on good data quality, proper infrastructure, employee training, and strong system support. These findings support the development of eFlow because it uses a structured digital system with AI features to improve task management, coordination, and decision-making in LEDIPO operations.

Process Mining Applications in Government Sector: A Systematic Literature Review

The study “Process Mining Applications in Government Sector: A Systematic Literature Review” explains how process mining is used to analyze digital records in government systems to improve efficiency and workflow management. It shows that government agencies generate large amounts of data, and this data can be used to identify delays, bottlenecks, and inefficiencies in processes. Through this, agencies are able to improve transparency, decision-making, and the overall delivery of public services. The literature also highlights that process mining is commonly applied in e-governance systems such as workflow management tools, business intelligence platforms, and government databases to support better monitoring and resource management. Despite its advantages, several issues were identified in the study, including poor data quality, outdated systems, lack of technical skills, cybersecurity risks, and resistance to change among employees. These challenges affect how well process mining can be implemented in government operations. These ideas are relevant to eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, since both focus on improving how government work processes are managed. However, while process mining focuses on analyzing past system data to improve workflows, eFlow is designed for real-time project

and task management to support daily LEDIPO operations. Overall, the literature shows that digital systems can greatly improve efficiency, transparency, and decision-making in government services. However, their success depends on proper data management, skilled users, and strong system infrastructure. These findings support the development of eFlow as a centralized system that improves task monitoring, coordination, and workflow efficiency within LEDIPO.

Predicting Public Sector Employee Burnout Using Big Data Analysis

The study “Predicting Public Sector Employee Burnout Using Big Data Analysis” focuses on how big data and predictive analytics can help identify early signs of employee burnout in public sector organizations. It explains that burnout is usually caused by heavy workloads, stress, and organizational pressure, which can negatively affect employee performance and service delivery. Data such as attendance records, workload distribution, performance reports, and surveys can be analyzed to detect these early warning signs. With the use of predictive analytics, organizations can better manage workloads, support employee well-being, and improve decision-making. It also allows agencies to respond early before problems affect productivity and service quality. However, concerns were also raised regarding data privacy, ethical use of employee information, lack of infrastructure, limited technical skills, and resistance from employees due to monitoring issues.

These findings relate to eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, since both use data-driven approaches to improve government operations. While the study focuses on predicting employee burnout, eFlow focuses on managing projects and tasks within LEDIPO using AI-based support.

The literature highlights that predictive and data-driven systems improve efficiency, decision-making, and employee support when properly implemented. However, they require accurate data, proper training, and strong privacy protection. These insights support eFlow as it integrates AI features to improve task management, coordination, and decision-making in a more organized and efficient LEDIPO system.

Digital Transformation in the Public Administrations: A Guided Tour for Computer Scientists

Digital transformation has become a central focus in modern public administration as governments continue to improve efficiency, transparency, and citizen engagement through technology. The study *Digital Transformation in the Public Administrations: A Guided Tour for Computer Scientists*, explores how public institutions adopt technologies such as artificial intelligence, cloud computing, big data analytics, automation tools, and e-governance systems to improve government operations and service delivery. It explains that these technologies help automate administrative processes, reduce paperwork, and make public services more accessible and efficient for citizens. Improvements in transparency and accountability are also emphasized in the literature, as digital systems allow government transactions and records to become more traceable and easier to monitor. In addition, data-driven tools support better decision-making by allowing agencies to analyze large volumes of information and respond more effectively to public needs. Despite these advantages, several challenges remain in implementing digital transformation in public administration. These include limited infrastructure, financial constraints, cybersecurity risks, resistance to change, and lack of technical skills among

employees. The study also highlights that technology alone is not enough, as successful implementation depends on leadership, training, and proper planning.

These findings are relevant to eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, since both focus on improving government efficiency through digital solutions. However, while the literature focuses on broad public administration systems, eFlow specifically addresses internal project and task management within LEDIPO. It replaces manual tracking and fragmented communication tools with a centralized and structured system designed for LEDIPO workflows. Overall, the literature confirms that digital transformation improves efficiency, transparency, and decision-making in government operations. However, its success depends on proper infrastructure, user readiness, and continuous training. These insights support the development of eFlow, which integrates artificial intelligence features to enhance task management, coordination, and workflow efficiency within the LEDIPO.

Artificial Intelligence and Machine Learning in Public Administration: A Systematic Literature Review

Artificial intelligence (AI) and machine learning (ML) have become important tools in modern public administration as governments shift toward more data-driven and automated systems. The study *Artificial Intelligence and Machine Learning in Public Administration: A Systematic Literature Review*, examines how these technologies are applied to improve governance, decision-making, and service delivery. It explains that AI can automate repetitive tasks, reduce human errors, and improve operational efficiency in government institutions. Machine learning systems are also highlighted for their ability to analyze large

datasets, identify patterns, and support predictive decision-making. These technologies improve government services through tools such as chatbots, digital assistants, automated response systems, and intelligent public service platforms that make government services faster and more accessible. Alongside these benefits, the literature also identifies several challenges. These include data privacy concerns, cybersecurity risks, algorithm bias, and ethical issues in decision-making. Other limitations include lack of infrastructure, limited technical expertise, and resistance from employees due to changes in workflow systems. The study emphasizes that AI systems must be implemented with proper policies, training, and ethical guidelines to ensure responsible use.

In relation to eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, the literature supports the use of AI in improving government operations. Both focus on using intelligent systems to improve efficiency and decision-making. However, the literature focuses on general AI applications in public administration, while eFlow applies AI specifically for task management, workflow support, and internal coordination within LEDIPO. The findings show that AI and machine learning improve efficiency, accuracy, and service delivery in government systems. However, their success depends on proper data handling, infrastructure, ethical safeguards, and user readiness. These findings support eFlow's use of AI features to enhance task management, improve coordination, and support better decision-making within LEDIPO.

The Use of Blockchain Technology in e-Government Services

The study “The Use of Blockchain Technology in e-Government Services” explains how blockchain can improve the efficiency, transparency, and security of e-government systems. It describes how government services have shifted from paper-based processes to digital systems that are faster and more efficient. Blockchain technology further strengthens these systems by providing secure, transparent, and tamper-proof records. The literature highlights that blockchain improves trust in government systems because all transactions are permanently recorded and cannot easily be changed. This makes it useful for applications such as document processing, identity management, approvals, and audit trails. It also strengthens accountability because all actions in the system can be traced and verified. In addition, blockchain supports digital workflow systems by reducing paperwork, improving coordination, and speeding up government processes. When combined with digital platforms, it helps improve efficiency and service delivery in public administration. These features are directly related to the proposed system, which aims to include a blockchain-based audit trail for tracking task updates, approvals, and system activities. The literature also notes that Artificial Intelligence can be integrated with blockchain and workflow systems to improve decision-making, workload management, and task distribution. Optimization methods such as Genetic Algorithms and Process Mining are also discussed as tools for improving scheduling, identifying inefficiencies, and analyzing workflows. However, challenges such as high implementation cost, technical complexity, lack of infrastructure, cybersecurity risks, and resistance to change are also identified. These issues show that successful implementation requires proper planning, training, and strong organizational support. Overall, the reviewed studies emphasize that

combining blockchain, AI, and intelligent workflow systems can greatly improve transparency, efficiency, and accountability in government operations. These findings strongly support the development of eFlow, as it integrates these technologies into a unified system designed to improve project and task management within LEDIPO.

A Genetic Algorithm for the Resource Constrained Multi-Project Scheduling Problem

The study entitled "A Genetic Algorithm for the Resource Constrained Multi-Project Scheduling Problem" by J.F. Gonçalves, J.J.M. Mendes, and M.G.C. Resende discussed the use of Genetic Algorithms in solving complex scheduling and resource allocation problems involving multiple projects with limited resources. The researchers developed an optimization approach that uses Genetic Algorithms and heuristic scheduling techniques to create efficient project schedules based on priorities, release dates, delay times, and operational constraints. The findings of the study revealed that the proposed algorithm effectively improved scheduling efficiency, optimized resource utilization, and reduced conflicts in multi-project environments. The computational results validated the effectiveness of Genetic Algorithms in handling resource-constrained scheduling problems and demonstrated their capability to improve operational performance. However, the researchers also noted that the effectiveness of Genetic Algorithms depends heavily on the quality of input parameters and constraints, and that poorly defined fitness functions may result in suboptimal scheduling outcomes. These findings suggest that proper configuration and parameter tuning are essential prerequisites for achieving reliable and accurate optimization results in real-world scheduling environments.

The literature is closely related to the proposed capstone project because it supports the implementation of a Genetic Algorithm optimization mechanism using the PyGAD library for intelligent task scheduling and workload balancing within LEDIPO. Similar to the research, the proposed system analyzes workload conditions, employee availability, task priorities, and deadlines to generate optimized task assignments. The identified limitations regarding parameter configuration further inform the development approach of the proposed system, ensuring that the PyGAD implementation is carefully configured with well-defined constraints suited to the operational requirements of LEDIPO.

METHODOLOGY

Technical Background

The research proponents gathered the necessary information and carefully studied various technologies, frameworks, and software tools that will be used in the development of the proposed system. Throughout the development process, the proponents will utilize different development technologies and tools to ensure that the system is efficient, responsive, secure, and user-friendly for all intended users. These technologies will also help improve the overall functionality, performance, and accessibility of the system across both web and mobile platforms. The proposed project, titled of eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration, is designed for LEDIPO employees, department heads, supervisors, and administrators. This project is an IT-related system that involves the use of different programming languages, frameworks, databases, and development tools commonly applied in modern web and

mobile application development. The system aims to centralize project and task management processes such as task assignment, progress monitoring, notifications, reporting, and collaboration into a single integrated platform, making workflow management more organized, efficient, and accessible within the local government unit.

In developing the system, the proponents will rely on several technological resources that will serve as the foundation of the project. These resources include front-end and back-end technologies, database management systems, mobile and web development frameworks, and supporting software tools that will ensure smooth system operation and efficient data management. Each technology will play an important role in the development process, contributing to the overall structure and functionality of the system, from interface design and real-time task monitoring to data processing, reporting, and system deployment.

Technologies to be Used

- **TypeScript** – will be used as the primary programming language for the mobile frontend, providing static type checking on top of JavaScript to improve code reliability, catch errors during development, and enhance overall maintainability, consistent with the web system's language choice.
- **React** – will be used as the main frontend library for building the web-based user interface, enabling the development of reusable components and efficient rendering through its virtual DOM.
- **React Native** – will be used as the main framework for building the mobile application, enabling the development of cross-platform apps for both Android and iOS from a single codebase using familiar React patterns and component-based architecture.

- **Expo (SDK 55)** – will be used as the development platform and toolchain built on top of React Native, simplifying the build process, providing a rich set of native device APIs, and enabling rapid development through Expo Go for instant testing on physical devices without requiring native build tools.
- **Vite** – will be used as the build tool and local development server for the web application, offering fast hot module replacement and optimized production builds for the React system.
- **React Navigation (v7)** – will be used as the primary navigation library for the mobile application, providing stack, tab, and drawer navigators to handle screen transitions, deep linking, and navigation state management across the app.
- **Tailwind CSS** – will be used as a utility-first CSS framework to style the web system with a consistent and responsive design, enabling rapid UI development without extensive custom CSS.
- **NativeWind (v4)** – will be used as the styling solution for the mobile application, bringing the same Tailwind CSS utility-class workflow used in the web system to React Native, ensuring design consistency and enabling rapid UI development with familiar class-based styling.
- **shadcn/ui** – will be used as the component library for the web application, built on top of Radix UI primitives and providing accessible and customizable components such as dialogs, dropdowns, tabs, and form elements.
- **Gluestack UI (v2)** – will be used as the component library for the mobile application, providing accessible and customizable UI components such as buttons,

inputs, modals, and bottom sheets built on top of NativeWind, serving as the mobile equivalent of the web system's shadcn/ui.

- **Recharts** – will be used to render interactive charts and data visualizations within the web dashboard, displaying project metrics and analytical data in a readable and organized format.
- **Victory Native (v41)** – will be used to render interactive charts and data visualizations within the mobile dashboard, displaying project metrics and analytical data in a performant, mobile-optimized format, serving as the mobile equivalent of the web system's Recharts.
- **React Hook Form (v7)** – will be used to manage form state and validation in the mobile application, the same library used in the web system, reducing re-renders and simplifying the handling of user input across all forms in the app.
- **Zod (v4)** – will be used alongside React Hook Form via `@hookform/resolvers` for schema-based form validation, providing TypeScript-first runtime validation that ensures data integrity before submission to the backend.
- **Zustand (v5)** – will be used as the global state management solution for the mobile application, handling shared application state such as active user sessions, project data, and AI chat history across screens in a lightweight and unopinionated manner.
- **Axios (v1)** – will be used as the HTTP client for making REST API requests to the shared FastAPI backend, handling chat completions, model management, and other server-side operations from the mobile application.
- **Firebase** – will be used for authentication and real-time database services, handling user login and securely storing configuration data and notifications.

- **Supabase/Supabase-js (v2)** – will be used to connect the mobile application to the shared Supabase backend, providing the same PostgreSQL database access and RESTful API used by the web system for storing and retrieving structured project management data.
- **React-Native-Firebase (v24)** – will be used as the native Firebase SDK for the mobile application, handling user authentication and NoSQL real-time database services, providing better performance than the web Firebase SDK by leveraging the native iOS and Android Firebase SDKs directly.
- **pdfjs-dist** – will be used in the web system to render and process PDF documents directly in the browser, enabling users to preview uploaded reports and project files without requiring server-side rendering.
- **React-Native-Pdf** – will be used to render PDF documents natively within the mobile application, allowing users to view uploaded project files and reports directly on their device, serving as the mobile equivalent of the web system's pdfjs-dist.
- **Python** – will be used as the backend programming language, powering the server-side logic responsible for workflow processing, analytics, intelligent task allocation, and backend services.
- **FastAPI** – will be used as the Python web framework for building the backend REST API, exposing endpoints for authentication, project management, notifications, reporting, analytics, and system monitoring with high performance and asynchronous support.

- **Uvicorn** – will be used as the ASGI server to run the FastAPI backend locally and in deployment environments with support for asynchronous request handling.
- **llama-cpp-python** – will be used as the inference engine for running GGUF-format large language models locally, enabling intelligent automation and AI-assisted features without depending on external AI API services.
- **HuggingFace Hub** – will be used to programmatically download and manage GGUF model files from HuggingFace repositories for AI-related system functionalities.
- **PyGAD** – will be used as the open-source Python library for implementing the Genetic Algorithm component of the system, handling intelligent workload allocation by optimizing task assignments based on personnel availability, workload balancing, deadlines, and task priorities through customizable fitness functions, crossover, mutation, and parent selection operators.
- **Polygon** – will be used as the blockchain infrastructure for implementing the system's immutable audit trail feature, providing a scalable and low-fee blockchain network to record and verify task-related logs and transactions, ensuring that system activities are tamper-proof, transparent, and permanently traceable.
- **EAS Build (Expo Application Services)** – will be used as the cloud build service for compiling and generating production-ready Android APK and iOS IPA files without requiring local native build environments such as Android Studio or Xcode.
- **Visual Studio Code** – will be used as the primary code editor for developing the mobile system, the same editor used for the web system, offering extensions for

React Native and Expo development, debugging tools, syntax highlighting, and integrated version control support.

Calendar of Activities

Project Timeline and Module Milestones This section establishes the systematic execution timeline and module milestones for the development and deployment of eFlow. To maintain an organized engineering workflow, the system lifecycle is split across its three core operational environments: the Administrator Web Command Center, the Department Head and Supervisor Web Dashboard, and the Employee-facing Web and Mobile Application with role-activated access controls.

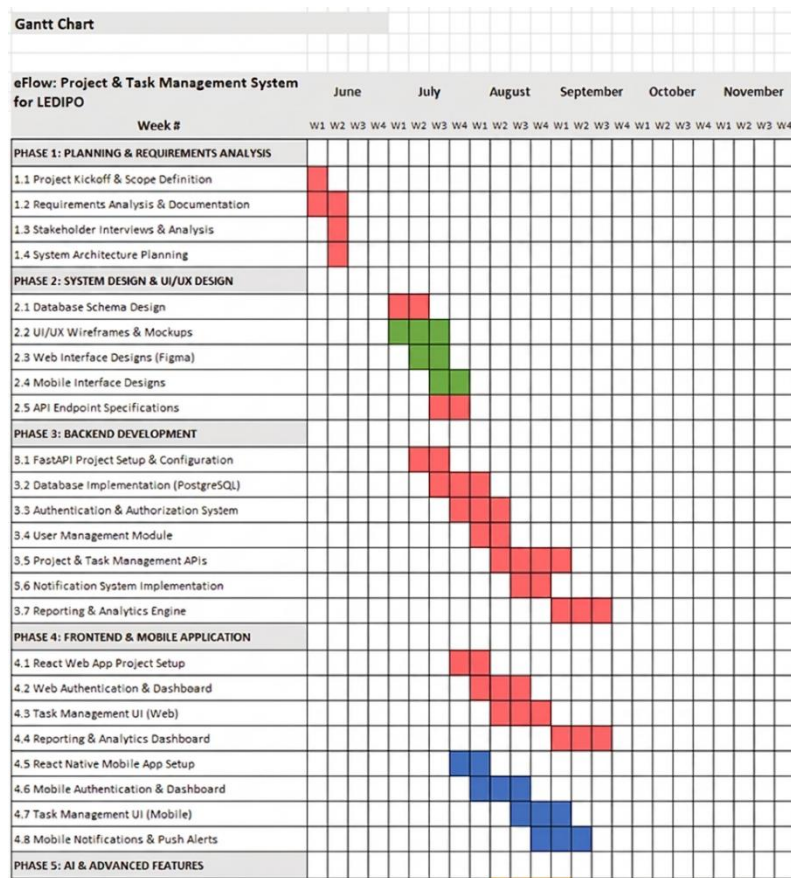
The project schedule covers a six-month development window spanning from June to November. This timeline maps development velocities, risk mitigation windows, and primary task assignments among the research proponents: Cahiyang, Denoy, Dichosa, and Manili. **Project Timeline Matrix (Gantt Chart)** The table below details the chronological Gantt chart schedule for the eFlow engineering lifecycle. Each operational month is partitioned into four blocks representing individual development weeks.

The development effort is organized across seven phases: Planning and Requirements Analysis, System Design and UI/UX Design, Backend Development, Frontend and Mobile Application Development, AI and Advanced Features Integration, Quality Assurance and Testing, and Deployment and Documentation. Task ownership is color-coded by proponent — **Ernst Livin Gabriel B. Cahiyang** serves as Lead Programmer and Applied AI Developer, **James Adrian A. Denoy** as Android Developer, **Andres U. Manili** as Designer and Blockchain Developer, and **Kurt Russell C. Dichosa** as QA Lead and

Genetic Algorithms specialist — ensuring accountability and parallel development across all system layers throughout the six-month timeline.

Project Timeline Matrix (Gantt Chart)

The table below details the chronological Gantt chart schedule for the eFlow engineering lifecycle. Each operational month is partitioned into four blocks representing individual development weeks.



PHASE 5: AI & ADVANCED FEATURES									
5.1 Genetic Algorithm Implementation (PyGAD)									
5.2 Intelligent Workload Allocation Module									
5.3 Process Mining Analysis Integration									
5.4 Blockchain Audit Trail Setup (Polygon)									
5.5 Blockchain Transaction Implementation									
5.6 AI-Assisted Task Recommendations									
PHASE 6: QUALITY ASSURANCE & TESTING									
6.1 Unit Testing (Backend)									
6.2 Integration Testing									
6.3 User Acceptance Testing (UAT)									
6.4 Performance & Load Testing									
6.5 Security & Penetration Testing									
6.6 Mobile App Testing (Android & iOS)									
PHASE 7: DEPLOYMENT & DOCUMENTATION									
7.1 Web Application Deployment									
7.2 Mobile Application Deployment (EAS)									
7.3 System Documentation & Manuals									
7.4 Training & Knowledge Transfer									
7.5 Project Presentation & Defense									

COLOR LEGEND

Ernst Livin Gabriel B. Cahiyani – Lead Programmer / Applied AI Developer / AI Process Mining

James Adrian A. Denoy – Android Developer

Andres U. Manili – Designer / Blockchain Developer

Kurt Russell C. Dichosa – QA Lead / Genetic Algorithms

Resources

This section specifies the hardware and software resources required to complete the development, testing, and implementation of **eFlow**.

Software Resources

- **Figma:** This serves as the primary UI/UX design tool for creating high-fidelity storyboards and interactive prototypes. It allows for the precise application of the STI color palette and layout testing before the development phase begins.
- **Visual Studio Code (VS Code):** This integrated development environment (IDE) functions as the central workspace for writing the source code. It supports the development of both the React Native + Expo mobile application and the React/Vite web dashboards through various debugging and syntax-highlighting extensions.

- **Firestore Console:** This web-based interface manages the backend infrastructure, including real-time data storage, user authentication, and cloud storage for uploaded project files and documents.

Hardware Resources

- **Development Laptops:** High-performance computers provide the processing power required to run the development IDEs, design software, and local server environments simultaneously.
- **Mobile Testing Devices:** A selection of Android and iOS smartphones enables the validation of the QR code scanning functionality and the "face-to-screen" visual verification process under various lighting conditions at campus events.

Requirement Analysis

The requirements analysis identifies the operational needs of the Local Economic Development and Investment Promotion Office of the proposed eFlow system workflows.

- **Who:** The primary participants include LEDIPO employees (task executors and collaborators), department heads and supervisors (project managers and approvers), administrators (system overseers and access managers), and system developers (maintainers and technology integrators responsible for AI, Process Mining, Genetic Algorithm optimization, and blockchain implementation).
- **What:** The core activities involve project creation, task assignment, workload monitoring, employee collaboration, workflow tracking, report generation, AI-assisted workload allocation, task prioritization, process analysis through Process Mining, and blockchain-based audit trail recording for transparency and accountability.

- **Where:** The system operates within the various departments and offices of the Local Economic Development and Investment Promotion Office and is accessible through web and mobile platforms using desktop computers, laptops, tablets, and smartphones connected to the internet. The technical environment includes React and React Native applications integrated with FastAPI, Supabase PostgreSQL, Firebase services, AI modules using llama-cpp-python, and Polygon blockchain infrastructure.
- **When:** The system is active during daily LEDIPO operations involving project planning, task delegation, workflow monitoring, reporting, and interdepartmental coordination. It is continuously used throughout active project cycles for monitoring deadlines, task progress, approvals, and project completion, while also providing anytime access to dashboards, analytics, notifications, and reports for authorized users.
- **How (Current Procedure):** The current manual and fragmented process involves several inefficient and repetitive operational steps:
 1. **Task Assignment:** Supervisors and department heads manually assign tasks through verbal instructions, spreadsheets, or messaging applications.
 2. **Monitoring:** Project tracking and progress updates are commonly managed using Microsoft Excel or Google Spreadsheets, making updates difficult to consolidate and monitor in real time.
 3. **Communication:** Coordination between employees is conducted through face-to-face meetings, Zoom meetings, text messaging, or Viber conversations across multiple communication channels.

4. Reporting: Employees manually submit updates and reports through chats, spreadsheets, or printed documents, causing delays, duplication of work, and inconsistent records.
5. Workflow Tracking: Monitoring approvals, deadlines, and completed tasks requires manual checking and follow-ups, resulting in inefficient coordination, limited transparency, and difficulty identifying workflow bottlenecks.

eFlow replaces these fragmented “how” procedures with a centralized and intelligent digital workflow management platform. Supervisors and administrators can create projects, assign tasks, monitor workflows, and validate outputs in real time, while employees can update task statuses, communicate, and submit deliverables directly within the system. Artificial Intelligence with Genetic Algorithm optimization assists in workload allocation and task prioritization, Process Mining analyzes workflow inefficiencies and bottlenecks, and blockchain-based audit trails securely record all task-related activities for transparency, accountability, and traceability within the Local Economic Development and Investment Promotion Office.

Requirements Documentation

This section establishes the formal agreement regarding the functional capabilities of the proposed of eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration. It defines the system’s core functionalities and describes the operational flow of the platform across its different user interfaces. The system is designed as a centralized task and project management platform that supports

real-time coordination, monitoring, and reporting within the Local Economic Development and Investment Promotion Office. It aims to replace manual and fragmented workflows with an integrated digital system accessible through both web and mobile applications.

1. SPECIFIC REQUIREMENTS

1.1 Functional Requirements

1.1.1 Login and Authentication Module

REQ001. The admin shall log in using a secure username and password.

REQ002. The employee/user shall register an account using official LEDIPO credentials and personal information.

REQ003. The employee/user shall log in using registered email and password credentials.

REQ004. The system shall display an error message for invalid login attempts.

REQ005. The system shall allow users to securely log out from the platform.

REQ006. The system shall support role-based access for employees, supervisors, and administrators.

1.1.2 Home Page and Dashboard Module

REQ007. The system shall display a centralized dashboard for ongoing projects and assigned tasks.

REQ008. The system shall provide navigation menus for Tasks, Projects, Reports, Notifications, and User Profile.

REQ009. The employee dashboard shall display assigned tasks, deadlines, and progress updates.

REQ010. The supervisor dashboard shall display department project summaries and employee task progress.

REQ011. The admin dashboard shall provide access to system-wide project monitoring and management tools.

REQ012. The system shall display announcements, reminders, and project updates on the homepage.

1.1.3 Project and Task Management Module

REQ013. The supervisor/admin shall create new projects and task assignments.

REQ014. The system shall allow supervisors to assign tasks to employees with deadlines and priorities.

REQ015. The employee/user shall view assigned tasks and project details.

REQ016. The employee/user shall update task progress and completion status in real time.

REQ017. The system shall display task status such as Pending, Ongoing, Completed, and Delayed.

REQ018. The supervisor/admin shall edit or update project and task details when necessary.

REQ019. The supervisor/admin shall archive or deactivate completed projects and tasks.

REQ020. The system shall maintain records of completed and ongoing tasks for monitoring purposes.

1.1.4 Collaboration and Communication Module

REQ021. The system shall allow users to post comments and updates on tasks and projects.

REQ022. The system shall support communication between employees and supervisors within the platform.

REQ023. The system shall notify users regarding task updates, comments, and project changes.

REQ024. The system shall display real-time updates related to assigned projects and activities.

1.1.5 Reporting and Analytics Module

REQ025. The system shall generate reports for project progress and task completion rates.

REQ026. The supervisor/admin shall view employee productivity and project performance summaries.

REQ027. The system shall provide downloadable reports for monitoring and evaluation purposes.

REQ028. The system shall display charts and analytics related to project and workflow performance.

1.1.6 Notification Module

REQ029. The system shall send notifications for newly assigned tasks.

REQ030. The system shall notify users regarding approaching deadlines and overdue tasks.

REQ031. The system shall send updates regarding project status changes and announcements.

REQ032. The system shall display reminders and alerts on both platforms.

1.1.7 User Profile Management Module

REQ033. The employee/user shall manage and update profile information.

REQ034. The system shall display user task history and assigned projects.

REQ035. The user shall securely update account credentials and passwords.

REQ036. The admin shall manage registered user accounts and permissions.

1.1.8 System Settings and Security Module

REQ037. The admin shall configure system settings and user access permissions.

REQ038. The system shall securely store user, task, and project information in the database.

REQ039. The system shall ensure compatibility with modern web browsers and mobile devices.

REQ040. The system shall maintain activity logs for monitoring user actions and system operations.

2. NON-FUNCTIONAL REQUIREMENTS

2.2.1 Operational Requirements

REQ041. The system shall operate on modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.

REQ042. The system shall support responsive design for desktop, tablet, and mobile devices.

REQ043. The system shall support simultaneous access for multiple users within the LEDIPO.

REQ044. The system shall provide stable operation with minimal downtime during normal usage.

REQ045. The system shall perform regular database backups to ensure data recovery and integrity.

REQ046. The system shall provide real-time synchronization of task updates across web and mobile platforms.

2.2.2 Security Requirements

REQ047. The system shall encrypt user passwords using secure hashing algorithms.

REQ048. The system shall allow access to projects, tasks, and reports only for authenticated users.

REQ049. The system shall restrict administrative functions to authorized admin accounts only.

REQ050. The system shall implement role-based access control (RBAC) for employees, supervisors, and administrators.

REQ051. The system shall securely store user information, task records, and project data within the database.

REQ052. The system shall maintain activity logs for monitoring and auditing purposes.

2.2.3 Performance Requirements

REQ053. The system shall load dashboard pages and task data within an acceptable response time under normal usage conditions.

REQ054. The system shall efficiently handle multiple task updates and notifications in real time.

REQ055. The system shall maintain consistent performance during simultaneous user access.

2.2.4 Cultural and Legal Requirements

REQ056. The system shall comply with the Data Privacy Act of 2012 and other applicable laws and regulations in the Philippines.

REQ057. The system shall use English as the default system language.

REQ058. The system shall ensure that project and organizational information are presented in a professional and government-appropriate manner.

3. TECHNICAL REQUIREMENTS

3.1 Front-End

- The system shall be developed using TypeScript for both web and mobile application development to improve code reliability and maintainability.
- The mobile application shall be developed using React Native with Expo SDK 55 to support cross-platform deployment for Android and iOS devices.
- The system shall use React Navigation v7 for screen navigation and routing within the mobile application.
- The user interface shall be styled using NativeWind v4 to provide a consistent and responsive design.
- The system shall utilize Gluestack UI v2 for reusable and accessible user interface components.

- The system shall use Victory Native v41 for charts, analytics, and dashboard visualizations.
- The system shall use React Hook Form v7 and Zod v4 for form handling and validation.

3.2 Back-End

- The system shall use Supabase with PostgreSQL as the primary database management system for storing and managing project, task, and user data.
- The system shall use Axios v1 for communication between the front-end and back-end services through REST API requests.
- The system shall support real-time data synchronization and notifications through integrated backend services.
- The system shall use React-Native-Firebase v24 for authentication and real-time services within the mobile application.

3.3 Security

- The system shall implement secure password encryption and authentication mechanisms to protect user accounts.
- The system shall apply role-based access control to restrict unauthorized access to sensitive system functions.
- The system shall ensure secure handling and storage of all projects, tasks, and user information.

3.4 Tools

- The system shall be developed using Visual Studio Code as the primary code editor and development environment.

- The system shall use EAS Build (Expo Application Services) for generating production-ready Android and iOS application builds.
- The system interface prototypes and UI/UX designs shall be created using design tools such as Figma and Canva.

Design of Software, System, Product, and/or Processes

This section presents the design of the proposed of eFlow: A Web and Mobile-Based e-Governance Project and Task Management System for the Local Economic Development and Investment Promotion Office with Artificial Intelligence Integration. The system is designed using modern software engineering principles and standard client-server architecture to ensure efficiency, scalability, usability, maintainability, and security. The overall design focuses on improving internal workflow management, communication, project monitoring, and task coordination within the Local Economic Development and Investment Promotion Office. The proposed system follows a client-server architecture, where the client-side is responsible for handling user interaction through both web and mobile interfaces, while the server-side manages business logic, authentication, database operations, notifications, and data processing. This architectural design allows real-time communication between users and the centralized database, enabling employees, supervisors, and administrators to access updated project and task information anytime and anywhere using internet-connected devices. To ensure organized development and modular functionality, the system is divided into several major components or modules. These include the Login and Authentication Module, Project and Task Management Module, Progress Monitoring Module, Notification Module, Reporting and Analytics Module, User Management Module, and Collaboration Module. Each module performs a

specific function but remains interconnected with other modules to support a unified and centralized workflow management platform. This modular structure also improves system maintainability and future scalability.

The system interface is designed with emphasis on usability, accessibility, and user experience. Since the system will be used by employees with different levels of digital literacy, the interface follows a simple and intuitive layout with organized navigation menus, clear task indicators, responsive dashboards, and easy-to-understand controls. The system supports both desktop and mobile devices, ensuring accessibility and convenience for users working inside or outside the office environment. The process flow of the system follows a structured sequence of operations. The process begins with user registration and secure authentication. Once logged in, supervisors and administrators can create projects, assign tasks, define priorities, and set deadlines. Employees can then receive assignments, update task progress, submit reports, and communicate through the system. Supervisors and administrators can monitor task completion, generate reports, analyze project performance, and receive real-time notifications regarding updates, delays, or completed activities. This workflow reduces dependency on manual tracking methods, physical meetings, and fragmented communication channels. The database design uses a relational database management structure to organize and manage system information efficiently. The database stores records related to user accounts, projects, tasks, reports, notifications, activity logs, and departmental information. Relationships between tables are properly established using primary and foreign keys to maintain data consistency, reduce redundancy, and ensure efficient retrieval of records. The database structure is also designed to support future expansion and integration of additional features if needed. The

system also incorporates security mechanisms to protect sensitive organizational data. Authentication, password encryption, role-based access control, and secured database connections are implemented to ensure that only authorized users can access specific functions and information within the platform. These security measures help maintain confidentiality, integrity, and accountability within the system. Overall, the design of the proposed eFlow system ensures that all software components, processes, and technologies work together effectively to provide a centralized, reliable, secure, and user-friendly e-governance project and task management platform. The system is designed not only to improve operational efficiency and productivity within LEDIPO but also to support its ongoing digital transformation initiatives toward more transparent, organized, and technology-driven governance.

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APPENDICES

APPENDIX A. RESOURCE PERSONS

Name: Rona Mira B. Lucañas

Profession: IT Program Head / Capstone Project Coordinator

Name of Institution: STI College Ormoc

Name: Cheryl P. Gallo

Profession: Administrative Officer

LGU ORMOC

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of
Ernst Livin Gabriel B. Cahiyang
Barangay Punta, Ormoc City
cahiyang.251639@ormoc.sti.edu.ph



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2021-2023	STI College Ormoc
High School	2017-2021	New Ormoc City National Highschool
Elementary	2011-2017	Punta Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2024 – Present	Web Developer	CeniXys
2025 - 2026	Backend Web Developer	Orynx

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT Club	Member
2023	CSS Club CIT	CSS Officer

SKILLS

SKILLS	Level of Competency	Date Acquired
Mobile Development	Proficient	2022
Web Development	Proficient	2023
System Design Architecture	Intermediate	2024
AI Development/Integration	Intermediate	2025

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
October 2025	Cyber Defense Exercise Qualifiers
September 2025	PayTaca Workshop
December 2024	National Service Training Program

Curriculum Vitae of
James Adrian A. Denoy
Barangay Tambulid, Ormoc City
denoy.265934@ormoc.sti.edu.ph



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2021-2023	STI College Ormoc
High School	2017-2021	New Ormoc City National Highschool
Elementary	2011-2017	Punta Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2024 - Present	Web Developer	CeniXys

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT & Sports Club	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Mobile Development	Intermediate	2022
Web Development	Proficient	2023
System Design Architecture	Intermediate	2024

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
October 2025	Cyber Defense Exercise Qualifiers
September 2025	PayTaca Workshop
December 2024	National Service Training Program

Curriculum Vitae of
Kurt Russell C. Dichosa
Barangay Punta, Ormoc City
dichosa.423829@ormoc.sti.edu.ph



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2021-2023	Ormoc City Regional Sports Academy
High School	2017-2021	New Ormoc City National Highschool
Elementary	2011-2017	Ormoc Sacred Heart Child Development Center

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT Club	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Mobile Development	Entry	2023
Web Development	Entry	2024
UI/UX Design	Entry	2022

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
April 2026	Bridging Research
September 2025	PayTaca Workshop
December 2024	National Service Training Program

Curriculum Vitae of
Andres U. Manili
Barangay Camp Downes, Ormoc City
manili.259643@ormoc.sti.edu.ph



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-2027	STI College Ormoc
Vocational/Technical	2021-2023	STI College Ormoc
High School	2017-2021	New Ormoc City National Highschool
Elementary	2011-2017	Doña Feliza Mejia Elementary Highschool

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2024	Youth Leader	New Life Church of Jesus Christ Philippines

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2026	IT Club	Member
2025	IT Club	Member
2024	IT Club	Member
2023	IT, Sports, Math and Science Club	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Mobile Development	Entry	2023
Web Development	Entry	2024
UI/UX Design	Entry	2022
Data Science	Entry	2024
System Design Architecture	Entry	2025

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
April 2026	Bridging Research
September 2025	PayTaca Workshop
December 2024	National Service Training Program